Tölvugrafík

Verkefni 3

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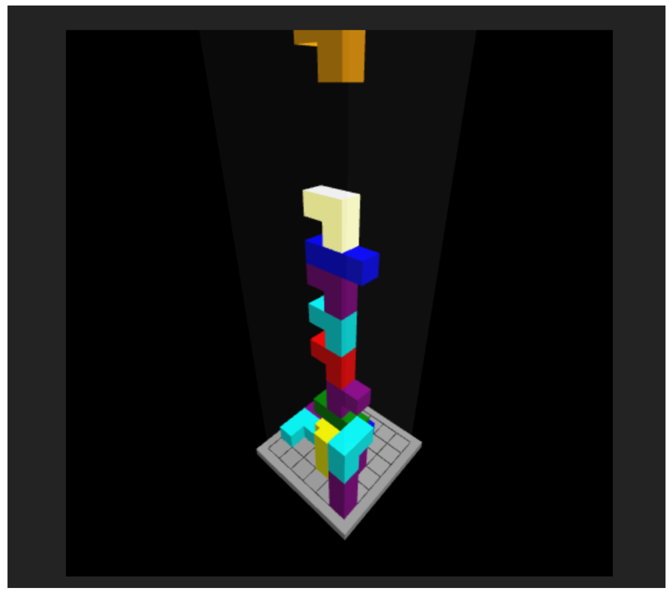
Github: https://github.com/JosephParaiso/Triis

Vefsíða: https://3dtetris.netlify.app/

Trominos



The trominos are created with three different blocks, either as a line or L-shaped, which are then grouped and can be rotated around the middle block. The trominos are also given random colours from a set array.

Collision

The logic for when a tromino collides with the ground was just based on world coordinates. But to keep track of collision with other trominos we created a three dimensional occupancy layer, each array corresponding to x, y or z, which keeps track of what cell is currently being occupied by a cube.

Stacking and consistency

After a tromino collides with either the ground or another tromino it locks into place. The program then ungroups each cube of the tromino, separating its individual cubes and recording them in a layers array according to their Y-position. At the same time, the corresponding cells in the three-dimensional occupancy array are updated to mark those positions as filled. This keeps both the visual scene and the logical grid representation consistent.

Deleting layers and blocks falling

Each frame, the program checks whether any layer is completely filled. This is done by iterating through the occupancy array layer by layer. If even a single cell in a layer is null, that layer is not considered full.

If a layer is fully occupied, it is deleted using the deleteLayer() function. This process removes the cubes from the scene and clears their entries from both the layers and occupancy arrays.

After deletion, the blockFall() function is called to move all layers above the cleared one down by one unit, updating both their world positions and their array entries. Since a layer is moved into the previously deleted layer, we must re-check the same layer after deletion, else we risk missing a completed layer.

Ghost block

The ghost block is a clone of the current falling tromino. It continuously mirrors the tromino's rotation as well as its X and Z position during movement.. How we did the logic of the positioning on the y level was we used the same collision logic we use for the trominos. Essentially every loop we drop down the ghost block until it detects collision.

The fun stuff

We decided to create an original song based on the classic tetris song which loops in the background while playing. We also added score. Each time a piece is placed, the user gains one point, and each time a layer is cleared, the user gains 100 points.